Ex. No: 2

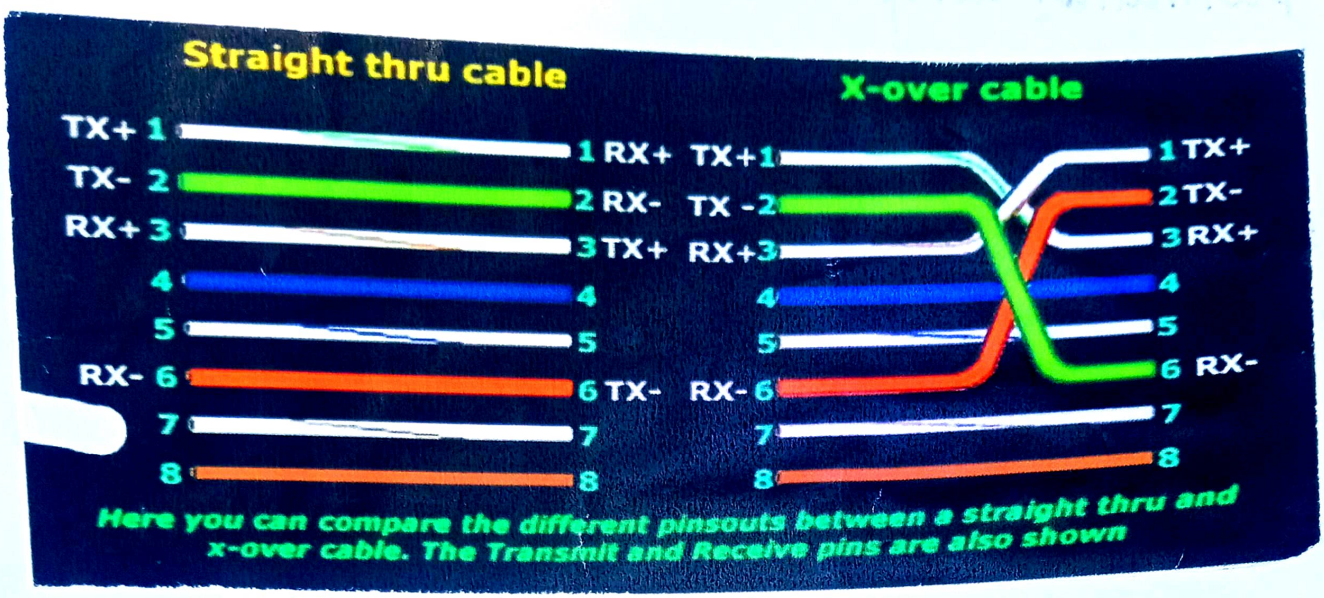
DATE: 23/7/26

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

AIM:

To study the different types of network cables used in computer networking and understand their purpose and applications.

cable type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / use
UTP	Category 3	10 bps	Advantages • cheaper in cost • Easy to install as they have a smaller overall diameter Disadvantages • More prone to (EMI) Electro-magnetic interface and noise.	10Base-T Ethernet
	Category 5	up to 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1Gbps		Fast Ethernet, Gigabit Ethernet
STP	Category 6, 6a	10Gbps	Advantages • Shielded • Faster than UTP. • Less susceptible to noise and interference.	Gigabit Ethernet, 10G Ethernet (55m) widely used in data centres

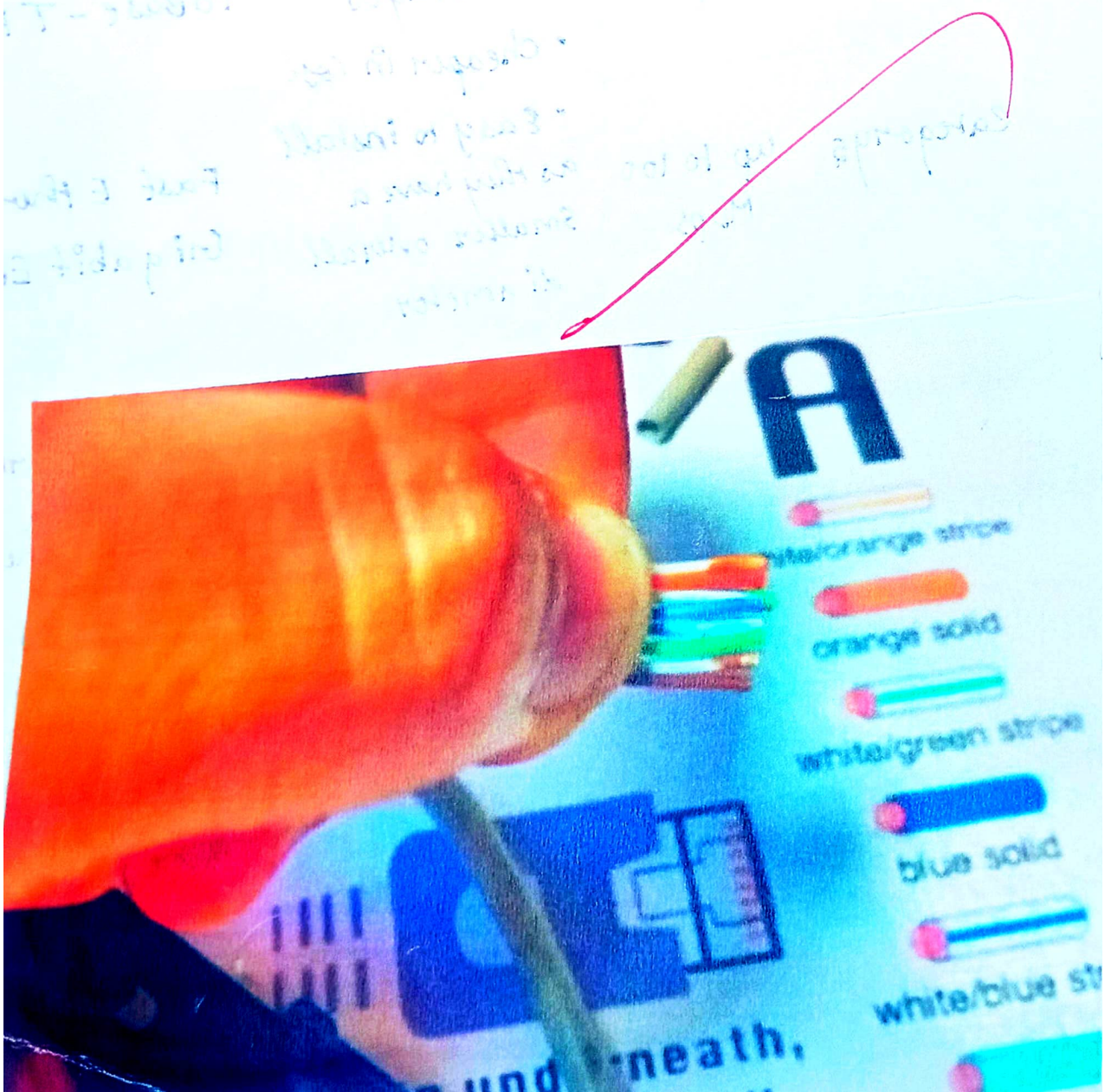

Straight thru cable
X-over cable
TX+ 1
TX- 2
RX+ 3
4
5
RX- 6
7
8
1 RX+
2 RX-
3 TX+
4
5
6 TX-
7
8
TX+1
TX -2
RX+3
4
5
RX-6
7
8
1 TX+
2 TX-
3 RX+
4
5
6 RX-
7
8
Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown


Clip underneath, pins facing you
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you
green solid
brown
blue
white

SSTP	Category 7	10 Gbps	Disadvantages • Expensive • Greater installation effort.	Gigabit Ethernet, 10G Ethernet. (100m)
Coaxial	RG-6 RG-59 RG-11	10-100 Mbps	• High bandwidth • Immune to interface • Low loss bandwidth • versatile Disadvantages • Limited distance • Cost • Size is bulky.	Speed of signal is 500m Television network High speed internet connections.
Fibre optics Cable	Single mode Multimode	100 Gbps	Advantages • High speed • High bandwidth • High security • Long distance Disadvantages • Expensive • Requires Skilled installers	• Maximum distance of fibre optics cable is around 100 meters

b) Make your own Ethernet Cross-over Cable / straight Cable


A
B


A
orange solid
blue solid

Tools and parts needed:

- Ethernet Cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
- A Crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs.
- Optional two plug shields.

Step 1:. To start construction of the device, begin by threading shields onto the cable.

Step 2:. Next, strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3:. After, you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. one end should be in arrangement A and the other in B.

Step 4:- Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order.

To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you, as shown.

Step 5:- Next, push the cable might in. The notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means that you stripped off too much shielding. Simply snip the cables back a little more.

Step 6:- After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

Step 7:- Lastly, repeat for the other end using diagram B (to make a crossover cables) / using diagram A (to make a straight through cable).

RESULT:-

Thus, different types of network cables were studied successfully, and straigh through and cross over Ethernet cables were prepared and tested successfully.